

Computer Vision 24 Credit Final

AIN SHAMS UNIVERSITY
FACULTY OF COMPUTER & INFORMATION SCIENCES
I-CREDIT HOURS PROGRAMS



Fall 2023 Module Code: CN6141 Module Name: Robotics Vision and Learning
Course Code: AIT 417 Time allowed: 2:00 Hrs.

Artificial Intelligence – Final Exam

The Exam Consists of Four Questions in FOUR Pages. Maximum Marks: 100 Marks 1 / 4

Question (1): (25 Marks)

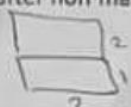
A) [9 Marks] State whether each of the following statements is either True or False.

1. If you want to segment an image to the best number of segments, the best method to do segmentation using clustering would be K-means. **False**
2. RANSAC is used to get rid of outliers in situations when there are wrong point correspondences. **True**
3. The main parameter of the Mean-shift algorithm is the initial window position. **Not Included**
4. The number of Visual words is the main parameter of classifying images using BOW. **True**
5. The task of image recognition is considered a sub-task of image verification. **True**
6. Classifying an object incorrectly is considered a false positive error in object detection. **True**
7. The YOLO model for object detection is more likely to produce multiple object responses in the image. **True**
8. While applying down-sampling of an image in the hierarchical LK algorithm, the process of image sharpening would help avoid aliasing. **Not Included**
9. Normalizing the SIFT descriptors in the direction of the dominant gradient orientation would allow the descriptors to be invariant to image translation. **False**

9- Normalizing SIFT descriptors in the direction of dominant gradient orientation makes them invariant to viewpoint and illumination changes but invariance to translation is achieved by using local key points and patches.

B) [16 Marks] Choose the correct answer

- In Harris detector, the decomposition of the eigenvalues λ_1 and λ_2 of the second moment matrix M reveals the existence of a corner when:
 - λ_1 and λ_2 are small
 - $\lambda_1 \gg \lambda_2$
 - λ_1 and λ_2 are large
 - $\lambda_2 \gg \lambda_1$
- You are building a 3-class object classification and localization algorithm. The classes are: pedestrian ($c=1$), car ($c=2$), motorcycle ($c=3$). What would be the label for the following image? Recall $y=[p_c, b_x, b_y, b_h, b_w, c_1, c_2, c_3]$

 - $y = [0, 0.3, 0.7, 0.3, 0.3, 0, 1, 0]$
 - $y = [1, 0.3, 0.7, 0.3, 0.3, 0, 1, 0]$
 - $y = [1, 0.3, 0.7, 0.5, 0.5, 1, 0, 0]$
 - $y = [0, 0.2, 0.4, 0.5, 0.5, 0, 1, 0]$
- Suppose two bounding boxes with sizes 2×1 and 2×3 are overlapping in a region of size 1×1 . The IoU between these two boxes will equal
 - 1/6
 - 1/9
 - 1/10
 - none of them
- Suppose you are using YOLO on a 19×19 grid, on a detection problem with 20 classes, and with 5 anchor boxes. During training, for each image you will need to construct an output volume y as the target value for the neural network; this corresponds to the last layer of the neural network. The dimension of this output volume will be
 - 19x19x45
 - 19x19x125
 - 19x19x500
 - none of them
- Suppose you run non-max suppression on the predicted boxes below. The parameters you use for non-max suppression are that boxes with probability ≤ 0.4 are discarded, and the IoU threshold for deciding if two boxes overlap is 0.5. How many boxes will remain after non-max suppression?
 

$\frac{2 \times 1}{2 \times 3} = \frac{2}{6}$

2- Example of the output vector:

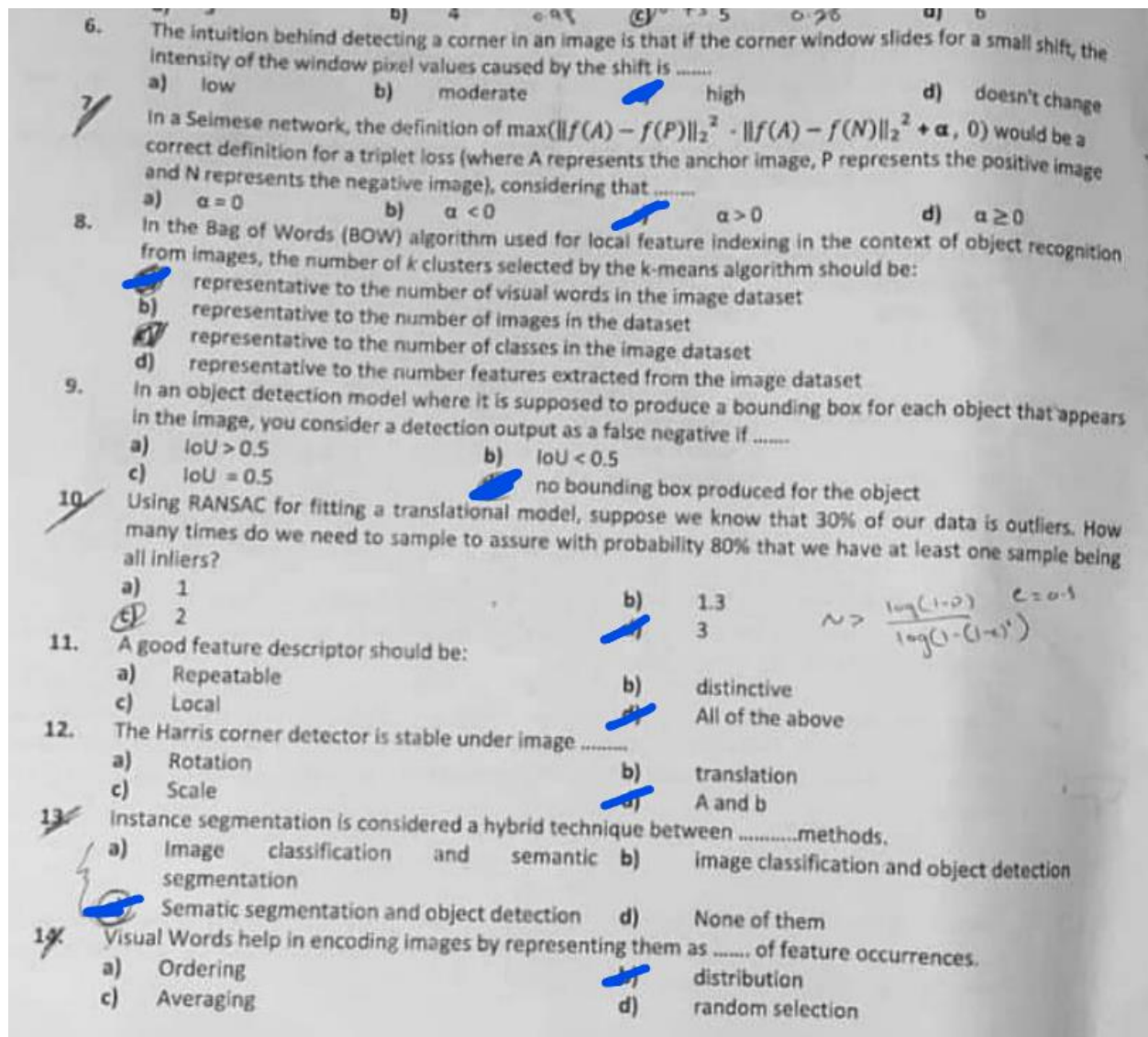


In our case: $PC = 1, C1 = 0, C2 = 1, C3 = 0$

$$3- IOU = \frac{\text{Area Of Overlap}}{\text{Area Of Union}} = \frac{1 \times 1}{2 \times 3 + 2 \times 1 - 1 \times 1} = 1/7$$

$$4- \text{Output volume} = (19 \times 19 \times (5 * 5 + 20))$$

5- NOT CLEAR ENOUGH :(



$$10- N > \frac{\log(1-p)}{\log(1-(1-e)^s)} = \frac{\log(1-0.8)}{\log(1-(1-0.3)^2)} = 2.39$$

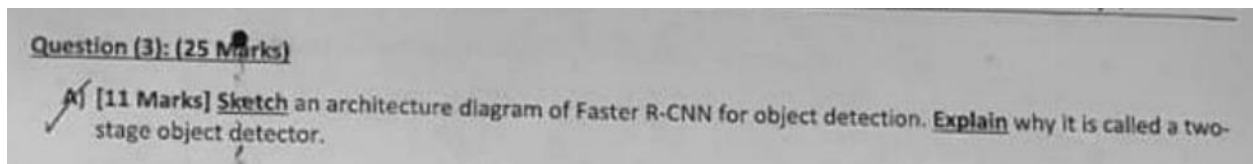
$$N = 3$$

II.

SIFT Algorithm STEPS:

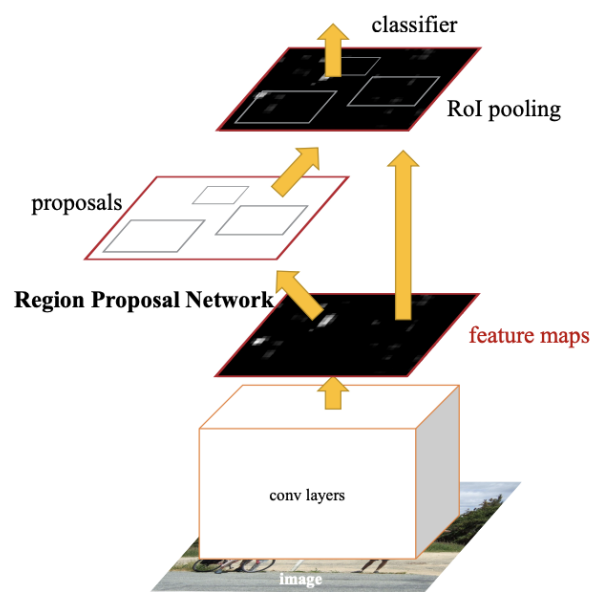
- Step 1: Constructing a scale space.
- Step 2: Laplacian of Gaussian approximation.
- Step 3: Keypoint localization.
- Step 4: Eliminate edges and low contrast regions.
- Step 5: Orientation Assignment.
- Step 6: Keypoint description.

Question 2: B: NOT INCLUDED



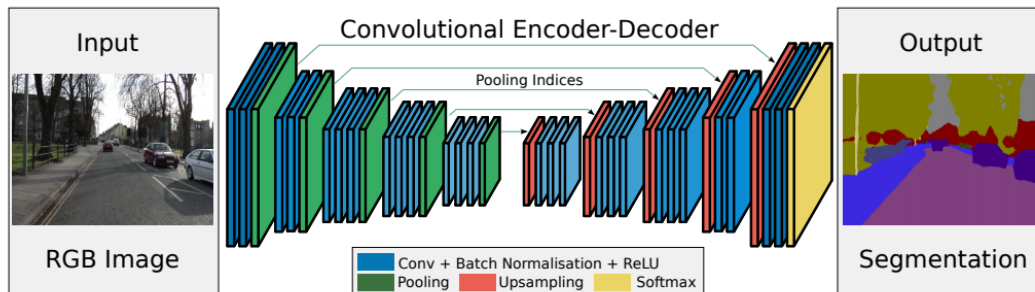
It's called a two-stage object detector because it performs object detection in two steps:

1. **Region Proposal:** The RPN generates regions of interest (ROIs) that are likely to contain objects.
2. **Object Detection:** The ROIs are refined and classified into objects or background.



B) [14 Marks] Sketch a general architecture of the Deconvolution network for semantic segmentation. Describe in detail the structure of the following: [6 marks]

- i. Input image. [2 marks]
- ii. Output image. [2 marks]
- iii. Final output layer of the network. [2 marks]
- iv. The network components and the function of each one. [2 marks]



I.

Input Image is RGB image ($H \times W \times c$)

II.

Output Image is image with same spatial dimensions

($H \times w \times \text{Number of channels equal number of classes}$)

III.

Usually, the final layer is a softmax layer.

($H \times w \times \text{Number of channels equal number of classes}$)

But I'm Not sure this is what the Dr means or not. ͇_(ツ)_/͇

IV.

Decoder Components (Right side of architecture):

1. Upsampling layers (Red blocks):
 - Increases spatial dimensions
 - Uses pooling indices from encoder
 - Restores resolution step by step
2. Convolutional blocks (Blue blocks):
 - Conv + Batch Normalization + ReLU
 - Refines features after upsampling
 - Gradually reduces feature channels
3. Final layers:
 - Yellow block: Softmax layer
 - Converts features to class probabilities
 - Produces final segmentation map

Question 4: NOT INCLUDED